
MarginOptimizer

B2B Dynamic Alternative Flight & Yield Management System

STRANDS AGENT AUTOMATED PNR ANALYSIS

CONFIDENTIAL · 2026

T A B L E O F C O N T E N T S

What This Document Covers

01 | The Cost Factor Extracting invisible profit from waiting tickets. **02 | The Solution Lifecycle** How the autonomous agent operates at 02:00 (Europe/Berlin). **03 | Infrastructure & Scaling** Processing thousands of items without triggering API bans (SQS). **04 | The Approval Loop (HITL)** Slack Webhooks and operator-side security. **05 | Development Roadmap** From local validation to cloud dashboard deployment.

THE COST FACTOR

Creating Money From Inefficiency

The Scenario: A package tour is sold today, departure in 3 months. The locked-in flight cost is €1,000. **The Opportunity:** In month 2 a competitor (or the same airline) drops an identical flight — same baggage, same window — to €800. Agencies miss it because a human cannot scan 10,000 PNRs every night. **The Solution:** MarginOptimizer runs one Haiku-class worker per PNR, autonomously, every night.

The ROI: $\sim 50,000 \text{ yearly bookings} \times 2\% \text{ profitable-hit rate} \times \text{€}40 \text{ average saving} = \geq \text{€}40,000 / \text{year}$ in pure recovered margin.

SYSTEM ARCHITECTURE

AWS Serverless High-Throughput Engine

SERVICE	FUNCTION IN THE ARCHITECTURE
AWS EventBridge	Nightly cron at 02:00 Europe/Berlin kicking the ingest Lambda
Amazon SQS	Queues 10,000 bookings with a DLQ, shielding Traffics from bursts
Lambda (Worker)	One Strands + Haiku 4.5 agent per PNR, max 10 concurrent
API Gateway	Receives Slack approval webhooks, HMAC-verified before routing
Aurora Postgres	Writes every approved swap to <code>yield_events</code> ; feeds QuickSight

STRANDSTOLING

The Margin Finders

- `use_traffics`
- **Target:** `/offers/{code}/alternativeFlights`
- **Execution:** Pings the Traffics spine to query identical alternative planes.
- `calculator_tool`
- **Execution:** Deterministically calculates (New Price - Old Price).
- **Integrity:** Prevents LLM mathematical hallucinations.
- `slack_tool`
- **Target:** Company operations channel.
- **Execution:** Dispatches interactive UI blocks (Approve/Reject) showing the found margin.

THE APPROVAL LOOP

Zero Unwanted Actions

The agent works entirely autonomously until a flight needs mutating. 1. The AI detects an €80 margin on Booking ID 4410. 2. It sends a Slack notification: "Revise flight to SunExpress to recover €80?" 3. The human operator clicks **[APPROVE]**. 4. An AWS Webhook bypasses AI and directly triggers Traffic `/bookings/modify`. 5. Absolute data safety achieved.

INVESTMENT & ROADMAP

Execution Strategy

Phase 1: Validation (0-1 Mo) Prove the specific prompt logic isolates profit perfectly locally. **Phase 2: Cloud Farm (1-2 Mo)** Deploy AWS SQS queues. Tweak rate-limit handling for 429s. **Phase 3: Operational Link (2-3 Mo)** Integrate the Slack Webhooks and the direct Mutation scripts. **Phase 4: Big Data Returns (4+ Mo)** Roll out Amazon QuickSight automated C-Level "Yield Dashboards".

MarginOptimizer

Autonomous, Invisible, Highly Profitable

CONFIDENTIAL & PROPRIETARY B2B YIELD
OPTIMIZATION AI.